

How disrupted spatial fluxes reshape food web structure along the river–sea continuum

Ismaël Lajaaiti, Eric Edeline, Anik Brindamour, Alain Danet

Table of contents

Abstract	1
Introduction	2
Freshwater ecosystems	2
Multiple stressors	2
Connectivity	2
Dams	3
Methods	3
Food web inference	3
Results	4
Discussion	4
Limitations	4
Supporting Information	5
Structure of the food web base	5
Model DAG	5

Abstract

Anthropic disturbances and ecosystem processes transcend the artificial boundaries between land, freshwater, and sea. While these systems are interconnected through species and material fluxes, they are often studied in isolation. In response, the concept of metaecosystem has emerged as a powerful framework to understand how fluxes among ecosystems shape their functions and dynamics. Disrupting these fluxes can have dramatic consequences that ripple across entire landscapes. In aquatic systems, dams and nutrient pollution are known to strongly affect fish communities. Yet, little is still known about how these effects propagate and potentially accumulate, through hydrographic basins and ultimately to the sea. Using bayesian spatio-temporal statistical models (R-INLA), we aim to analyse existing data on food-web structure across France, combined with environmental variables (dams, nutrients, ...), to quantify how breaking or increasing fluxes among systems disturbs food-web structure locally and how these disturbances spread throughout the metaecosystem.

Keywords: *foodweb ecology, stream network, spatial autocorrelation, bayesian statistics.*

Introduction

While patterns in community change within river network have been well studied, these studies have mostly focus on specise richness (Bonnaffé et al. 2024), or roughly defined functional composition (Vannote et al. 1980).

Freshwater ecosystems

- The River Continuum Concept has shaped the freshwater ecology research by links physical gradients along rivers to predictable changes in community structure, energy sources, and ecosystem functioning (Vannote et al. 1980; Doretto, Piano, and Larson 2020).
- Serial Discontinuity Concept (Ward and Stanford 1995).
- Community richness and hence food web structure is expected to be driven by nutrient availability (Ho, Altermatt, and Carraro 2023).

Multiple stressors

- Synergistic and antagonistic effects of multiple stressors (Jackson, Pawar, and Woodward 2021; Orr et al. 2024).
- Temperature can interact with nutrients and impact food web structure in a nonlinear way Binzer et al. (2012).
- (Danet et al. 2021; Bonnaffé et al. 2024).

Connectivity

- Small dispersal stabilizes through spatial structure and asynchrony (LeCraw, Kratina, and Srivastava 2014; McCann, Rasmussen, and Umbanhowar 2005; Pillai, Gonzalez, and Loreau 2011).
- Prey switching of mobile top predators can stabilize connected food webs through ‘spatial coupling’ (LeCraw, Kratina, and Srivastava 2014; McCann, Rasmussen, and Umbanhowar 2005; Pillai, Gonzalez, and Loreau 2011).
- At high dispersal/connectivity the stabilizing effect wanes as strongly connected patches behave as a single patch (LeCraw, Kratina, and Srivastava 2014; McCann, Rasmussen, and Umbanhowar 2005).
- Studies between pairs of the following connectivity, complexity and stability are common. However, studies investigating how these are interconnected are rare (LeCraw, Kratina, and Srivastava 2014).
- Connectivity has 4 dimensions: longitudinal, lateral, vertical and temporal (Grill et al. 2019).

- We identified five pressure factors to represent the main human interferences within the four dimensions of fluvial or river connectivity (Grill et al. 2019):
 - (1) river fragmentation (longitudinal);
 - (2) flow regulation (lateral and temporal);
 - (3) sediment trapping (longitudinal, lateral and vertical);
 - (4) water consumption (lateral, vertical and temporal);
 - (5) infrastructure development in riparian areas and floodplains (lateral and longitudinal).
- In freshwater system, fish dispersal is limited. A meta-analysis found that the mean dispersal kernel is around 10km (Comte and Olden 2018).

Dams

- Decrease river connectivity / reduce migration (Dean et al. 2023; He et al. 2024; Pedersen et al. 2012).
- Reservoir, lotic from lentic conditions, can result in hypoxic conditions. Favor phytoplanktons and macrophytes (Ding et al. 2026; Dean et al. 2023).
- Reservoir increase residence time of nutrients upstream and decrease the downstream output, which can result in a decrease in primary productivity.
- Hydropeaking - that is, the sudden release of large water to answer the higher electric demand - can harm riverine community due to sudden change in water level and water flow.
- Cumulative impact of dams can reduce the nutrient input in deltas (Dean et al. 2023).

Methods

Food web inference

To infer local food web structure, we used a metaweb approach. That is, we build a web compiling the potential feeding interactions between all species of the dataset. Importantly, our metaweb takes into account intra-specific variation in species diet. We did so by defining trophic interactions at different life stages for each species.

Metaweb nodes

The nodes of our metaweb represent a species at a given lifestage. Specifically, fish species were divided into nine body class sizes evenly distributed. Each body size class within a given species is what we call a ‘trophic species’ assuming that conspecifics of the same body size class share similar trophic interactions because trophic interactions are largely determined by predator–prey body size ratio in freshwater ecosystems (Brose, Williams, and Martinez 2006).

In addition of fish species, seven basal nodes (resources) were added at the base of the metaweb: detritus, biofilm, phytoplankton, zooplankton, macrophytes and zoobenthos. We represent the basal food web structure in the Appendix (see Figure S1). We assumed that these resources were present at all sampling sites.

Infer missing fish lengths

Most length of fishes are measured but not all. Yet, we need individual fish length to determine their diet based on their ontogeny. We infer missing fish size using information on the given fishing operation. When there are more than three observations of the fish species, we use a truncated normal distribution (package `truncdist`; Nadarajah and Kotz (2006)). The mean and variance are given by the modes of the observed distribution of the fishing operation. The minimum and maximum of the distribution are given by the minimal and maximal length observed on the entire data set for that species. We do so to avoid inferring unrealistic lengths.

Get fish diet

We use fishbase as a primary source to extract fish diet at their different life stages, to get on which basal categories species feed on and whether they are piscivorous or not. We detail in the Supporting Information how made the correspondance between our categories and the one of fishbase. We add a larvae stage when missing, assuming that fish larvae feed on zooplankton (Dabrowski 1984).

The metaweb

Subsampling the metaweb to get local food webs

Results

Discussion

Limitations

Regarding food web structure, we have simplified the base of the web and exclude macro-invertebrates. This choice naturally emphasizes the functional role of fishes in shaping food web structures. Yet, it would be valuable to include macro-invertebrates within the food web to get a more exhaustive picture of how aquatic community respond to stressors along the river-sea continuum.

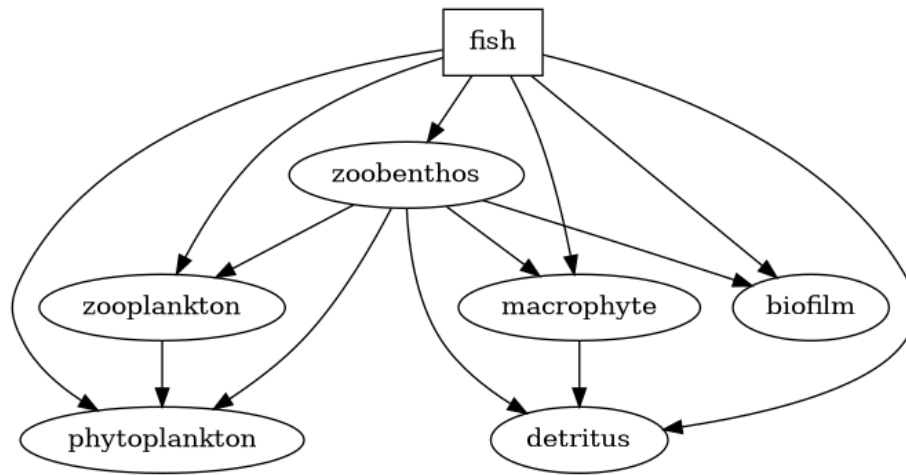


Figure S1: Assumed structure of the food web base.

Supporting Information

Structure of the food web base

Model DAG

Put model DAG here using targets.

- Binzer, Amrei, Christian Guill, Ulrich Brose, and Björn C. Rall. 2012. “The Dynamics of Food Chains Under Climate Change and Nutrient Enrichment.” *Philosophical Transactions of the Royal Society B: Biological Sciences* 367 (1605): 2935–44. <https://doi.org/10.1098/rstb.2012.0230>.
- Bonnaffé, Willem, Alain Danet, Camille Leclerc, Victor Frossard, Eric Edeline, and Arnaud Sentis. 2024. “The Interaction Between Warming and Enrichment Accelerates Food-Web Simplification in Freshwater Systems.” *Ecology Letters* 27 (8): e14480. <https://doi.org/10.1111/ele.14480>.
- Brose, Ulrich, Richard J. Williams, and Neo D. Martinez. 2006. “Allometric Scaling Enhances Stability in Complex Food Webs.” *Ecology Letters* 9 (11): 1228–36. <https://doi.org/10.1111/j.1461-0248.2006.00978.x>.
- Comte, Lise, and Julian D. Olden. 2018. “Fish Dispersal in Flowing Waters: A Synthesis of Movement- and Genetic-Based Studies.” *Fish and Fisheries* 19 (6): 1063–77. <https://doi.org/10.1111/faf.12312>.
- Dabrowski, K. 1984. “The Feeding of Fish Larvae : Present « State of the Art » and Perspectives.” *Reproduction Nutrition Développement* 24 (6): 807–33. <https://doi.org/10.1051/rnd:19840701>.
- Danet, Alain, Maud Mouchet, Willem Bonnaffé, Elisa Thébault, and Colin Fontaine. 2021. “Species Richness and Food-Web Structure Jointly Drive Community Biomass and Its Temporal Stability in Fish Communities.” *Ecol-*

- ogy Letters* 24 (11): 2364–77. <https://doi.org/10.1111/ele.13857>.
- Dean, E. M., Dana M. Infante, Hao Yu, Arthur Cooper, Lizhu Wang, and Jared Ross. 2023. “Cumulative Effects of Dams on Migratory Fishes Across the Conterminous United States: Regional Patterns in Fish Responses to River Network Fragmentation.” *River Research and Applications* 39 (9): 1736–48. <https://doi.org/10.1002/rra.4173>.
- Ding, Na, Zunyi Xie, Xiaoming Wang, Zongming Wang, Miao Li, Beiming Cai, Qinghe Zhao, et al. 2026. “Freshwater Biodiversity Impacts of Global Hydropower Dams.” *Communications Earth & Environment*, February. <https://doi.org/10.1038/s43247-026-03263-y>.
- Doretto, Alberto, Elena Piano, and Courtney E. Larson. 2020. “The River Continuum Concept: Lessons from the Past and Perspectives for the Future.” *Canadian Journal of Fisheries and Aquatic Sciences* 77 (11): 1853–64. <https://doi.org/10.1139/cjfas-2020-0039>.
- Grill, G., B. Lehner, M. Thieme, B. Geenen, D. Tickner, F. Antonelli, S. Babu, et al. 2019. “Mapping the World’s Free-Flowing Rivers.” *Nature* 569 (7755): 215–21. <https://doi.org/10.1038/s41586-019-1111-9>.
- He, Fengzhi, Christiane Zarfl, Klement Tockner, Julian D. Olden, Zilca Campos, Fábio Muniz, Jens-Christian Svenning, and Sonja C. Jähnig. 2024. “Hydropower Impacts on Riverine Biodiversity.” *Nature Reviews Earth & Environment* 5 (11): 755–72. <https://doi.org/10.1038/s43017-024-00596-0>.
- Ho, Hsi-Cheng, Florian Altermatt, and Luca Carraro. 2023. “Coupled Biological and Hydrological Processes Shape Spatial Food-Web Structures in Riverine Metacommunities.” *Frontiers in Ecology and Evolution* 11 (June). <https://doi.org/10.3389/fevo.2023.1147834>.
- Jackson, Michelle C., Samraat Pawar, and Guy Woodward. 2021. “The Temporal Dynamics of Multiple Stressor Effects: From Individuals to Ecosystems.” *Trends in Ecology & Evolution* 36 (5): 402–10. <https://doi.org/10.1016/j.tree.2021.01.005>.
- LeCraw, Robin M., Pavel Kratina, and Diane S. Srivastava. 2014. “Food Web Complexity and Stability Across Habitat Connectivity Gradients.” *Oecologia* 176 (4): 903–15. <https://doi.org/10.1007/s00442-014-3083-7>.
- McCann, K. S., J. B. Rasmussen, and J. Umbanhowar. 2005. “The Dynamics of Spatially Coupled Food Webs.” *Ecology Letters* 8 (5): 513–23. <https://doi.org/10.1111/j.1461-0248.2005.00742.x>.
- Nadarajah, Saralees, and Samuel Kotz. 2006. “R Programs for Truncated Distributions.” *Journal of Statistical Software* 16 (August): 1–8. <https://doi.org/10.18637/jss.v016.c02>.
- Orr, James A., Samuel J. Macaulay, Adriana Mordente, Benjamin Burgess, Dania Albin, Julia G. Hunn, Katherin Restrepo-Sulez, et al. 2024. “Studying Interactions Among Anthropogenic Stressors in Freshwater Ecosystems: A Systematic Review of 2396 Multiple-Stressor Experiments.” *Ecology Letters* 27 (6): e14463. <https://doi.org/10.1111/ele.14463>.
- Pedersen, M. I., N. Jepsen, K. Aarestrup, A. Koed, S. Pedersen, and F. Økland. 2012. “Loss of European Silver Eel Passing a Hydropower Station.” *Journal of Applied Ichthyology* 28 (2): 189–93. <https://doi.org/10.1111/j.1439-0426>.

[2011.01913.x](#).

- Pillai, Pradeep, Andrew Gonzalez, and Michel Loreau. 2011. "Metacommunity Theory Explains the Emergence of Food Web Complexity." *Proceedings of the National Academy of Sciences* 108 (48): 19293–98. <https://doi.org/10.1073/pnas.1106235108>.
- Vannote, Robin L., G. Wayne Minshall, Kenneth W. Cummins, James R. Sedell, and Colbert E. Cushing. 1980. "The River Continuum Concept." *Canadian Journal of Fisheries and Aquatic Sciences* 37 (1): 130–37. <https://doi.org/10.1139/f80-017>.
- Ward, J. V., and J. A. Stanford. 1995. "The Serial Discontinuity Concept: Extending the Model to Floodplain Rivers." *Regulated Rivers: Research & Management* 10 (2-4): 159–68. <https://doi.org/10.1002/rrr.3450100211>.